Recommendations for Fall Creek Monitoring Plan to address Sediment and Soluble Reactive Phosphorus Concentrations

TO: Cornell University, BEE 4270, New York State Department of Health

FROM: ToMaoz Engineering - Tomás Schmieder & Maoz Bizan

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1. Executive Summary

Fall Creek is an important drinking water source for Cornell University and a major tributary to Cayuga Lake, but it is affected by high sediment and phosphorus levels, especially during storms. Monitoring data from the Community Science Institute (CSI) show that most sediment comes from stream bank erosion, with the Varna Cliffs being the largest source in the watershed. Sediment levels increase downstream and rise sharply during high-flow events. High phosphorus levels are mainly linked to agricultural areas, where some monitoring sites show concentrations far above the watershed average. A phosphorus mass balance indicates that manure already supplies enough phosphorus for crops, while added fertilizer creates an unnecessary surplus that increases runoff risk. This report recommends focusing on the largest and most controllable threats to water quality. Sediment inputs should be reduced by structurally stabilizing the Varna Cliffs and eroding stream banks in Virgil Creek, and by improving monitoring around erosion hotspots and during storms. Phosphorus reductions should focus on eliminating or reducing purchased fertilizer while keeping manure applications, which can lower phosphorus runoff without hurting crop yields. Incentive-based monitoring programs are recommended to encourage farmer participation. These actions provide a practical way to improve water quality in the Fall Creek watershed, protect Cornell's drinking water supply, and ensure ecological stability in the watershed.

2. Introduction

This report establishes recommendations for a new watershed management plan for the Fall Creek watershed in Cayuga County, Cortland County, and Tompkins County, NY, which includes Fall Creek and Virgil Creek. First, current challenges facing the watershed are examined alongside approaches to managing them. Following this, new approaches to management in the watershed are suggested, aiming to reduce the impact of issues facing the watershed. This report will provide recommendations to address the concerns of the Cornell Filtration Plant and the Steward of Cayuga Lake. Adopting the recommendations outlined in this report into Fall Creek's management plan will ensure that the interests of all stakeholders in the watershed are addressed.

Fall Creek currently faces two main challenges: sedimentation and phosphorus input. These challenges originate from an array of natural and anthropogenic sources, and adversely impact drinking water, agriculture, recreation, and other activities within the Fall Creek watershed. This report recommends updates to the current Fall Creek management plan to address sedimentation and soluble reactive phosphorus concentrations within the watershed. Furthermore, it recommends updates to the current monitoring plan by the Community Science Institute (CSI) to ensure that the report's recommendations are effective at improving watershed management. Finally, it evaluates the reliability of artificial intelligence (AI) and large-language models (LLMs) at producing an effective watershed management plan. The recommendations in this report are based on analysis of data available from publicly available sources listed at the end of this report, and fieldwork within the watershed.

3. Analysis of the Issues

3.1. Sedimentation

Sedimentation in the Fall Creek watershed arises from distinct, measurable sources that vary geographically and by flow conditions. Current monitoring data collected by CSI reveals consistent patterns:

- Suspended sediment concentrations rise significantly from upstream to downstream;
- High flow events intensify sediment inputs;
- Specific geographic features act as major sediment sources.

Understanding how these patterns unfold within the watershed is essential to identifying effective management interventions.

Both Fall Creek and Virgil Creek show a general increase in suspended sediment concentrations as they approach their intersection in Freeville:

- In Virgil Creek, concentrations roughly double between Owego Hill Road (3.12 mg/L) and Freeville (6.94 mg/L) during baseflow;
- In Fall Creek, baseflow concentrations increase from 5.43 mg/L to 9.89 mg/L, a similar doubling.

Both streams demonstrate a combination of gradual, consistent increases in sediment concentrations as the stream flows and sudden spikes at locations along the stream. This suggests that sedimentation is caused by a combination of diffuse non-point sources and geographically identifiable point sources.

Periods of high flow, such as stormflows, cause higher sediment concentrations in the watershed. Concentrations in Virgil Creek rise tenfold, while in Fall Creek they are 30 times higher than during base flow periods. These increases suggest that high flow conditions are a strong driver of increased sedimentation in the watershed. High flow exacerbates conditions such as bank erosion and sediment runoff, suggesting that these phenomena are the culprits for the majority of sedimentation. Base flow sedimentation is much lower than high flow, so any management solutions should target the increases that happen during periods of high flow.

Each creek has at least one major identifiable point source of sedimentation. For Virgil Creek, this occurs downstream of the Dryden Village Sewage plant, just upstream of the Spring House Road Bridge measurement site: at this portion of the stream, there is a heavily meandering area where the creek actively erodes the banks (Figure 1). During times of peak flow, such as during storms, the largest jump in sedimentation occurs upstream of the Spring House Road Bridge measurement site and downstream of the Dryden Lake Outlet site, coinciding with the heavily meandering area. The dramatic increase – from 18.7 to 102.77 mg/L – suggests that erosion of the banks is responsible for most of this sedimentation in Virgil Creek. Downstream of the meandering area concentrations of sediment decrease, suggesting that most of the sediment in Virgil Creek comes from this point source.

In Fall Creek concentrations of sediment rise sharply and continue to increase, following a different pattern than Virgil Creek. Fall Creek faces a stark jump in sediment concentration between Freeville and the Freese Road Bridge, and a continued increase in concentrations downstream of the Freese Road Bridge. This jump coincides with the Cliffs of Varna, which are located along the stream's banks between these two measurement sites (figure 2). The cliffs are steep and visibly erode, releasing soil into Fall Creek and explaining the jump. The cliffs are located continuously along the edges of the stream and are unstable throughout their course, explaining the steadily increasing concentrations of

sediment as the stream flows. Furthermore, several areas within Fall Creek appear to be extremely shallow or largely bare soil (figure 3). These areas are likely also responsible for erosion and sediment deposition into the stream, contributing to the sedimentation issue.

Nearly half of sediment in the watershed originates from bank erosion during baseflow periods, and this proportion increases during storm events. This is calculated by subtracting the sediment concentration at the start of the streams from that at the end of the streams, and dividing this by the final sediment concentration. Full results are detailed in table 1. Accordingly, sedimentation is a major source of suspended solids in the watershed, and this largely results from bank erosion especially during storm events. Runoff of sediment is proportional to the watershed area of each stream. To calculate the total suspended solid concentration downstream of their junction, the following equation can be applied:

$$Average\ TSS\ concentration\ below\ the\ junction = \frac{Virgil\ Creek\ Drainage\ Area * Virgil\ Creek\ Concentration + Fall\ Creek\ Drainage\ Area * Fall\ Creek\ Concentration}{Total\ Drainage\ Area}$$

According to values available from the CSI website for each stream, the average total suspended solids concentration can be calculated for base flow conditions as well as storm conditions (table 2). During high flow conditions, the concentration is over seven times higher than during base flow conditions. The geographic and temporal patterns of sedimentation in the Fall Creek watershed illustrate the complexities of the issue: a mix of point and non-point sources, as well as strongly increased sedimentation during storm periods, which must be accounted for when revising the watershed management plan.

3.2. Phosphorus

Soluble reactive phosphorus (SRP) due to nutrient loading is a priority issue for Ithaca's water supply protection. Large amounts of SRP can directly fuel algal growth and eutrophication, which can compromise water treatment and quality. The Fall Creek and Virgil Creek watersheds have significant agricultural land use, with pastures and cultivated crops making up a large percentage of the land cover. Agricultural runoff containing nutrients from manure and fertilizers is a major source of phosphorus.

Figure 4 and 5, show the Community Science Institutes observations of SRP in monitoring locations on Fall Creek and Virgil Creek. It is notable that the SRP in two monitoring locations around Fall Creek are incredibly greater than the average. While all other monitoring locations have an average

of 14 ug/L of SRP the locations of Champlin Road Culvert and Tillotson Road Culvert have values of 394.81 ug/L and 543.5 ug/L respectively. Notably when looking at these areas on Google Maps the area surrounding these two monitoring locations are largely agricultural lands and directly on the other side of Fall Creek are two large CAFOs (concentrated animal feeding operations), however, it is unclear if these are related as the monitoring locations directly downstream of the CAFOs have minimal SRP levels.

It is also notable that when viewing the DWSP2s map of critical area, Tillotson Road Culvert is 0.60 miles away from the closest point to the creek, while the critical area of DWSP2s map only outlines around 0.18 miles in either direction of the creek. This is important as the critical area noted by the DWSP2 does not include the two greatest contributors of SRP based on CSIs findings.

The application of fertilizer to agricultural lands surrounding the watershed make up a large contributor to the increase in SRP in water sources. This is seen in Godber et al. (2024) where Table 2 shows the amounts of phosphorus from different sources in Tompkins, Cortland, and Cayuga county. When completing a mass balance analysis on these three Counties it was found that phosphorus accumulation is happening at great lengths due to the combined inputs from livestock manure (641,850 kg) and purchased fertilizers (260,700 kg) which exceeds the crop removal rates (635,250). This creates a watershed wide surplus of 8.10 kg P per hectare annually, with the situation being particularly present in harvested cropland areas where Phosphorus additions reach 1,886 kg per hectare. Cayuga County was found to have the most critical issue with a surplus of 15 kg P hectare, followed by Cortland and Tompkins Counties with surpluses of 9 and 6 kg P per hectare respectively. This persistent over-application of phosphorus, driven by the dual use of manure and commercial fertilizers on the same fields, leads to soil phosphorus saturation and increased risk of phosphorus runoff into Fall Creek, where the excess nutrients, particularly in the form of SRP can contribute to a decrease in water quality.

4. Use of A.I.

The A.I. generated watershed management plan differs from this report with the main difference being the lack of evidence. While this report focuses precisely on two data-drive issues, sedimentation and phosphorous loading, using data from CSI to identify specific problem locations like the Varna Cliffs

and sections of Virgil Creek, the A.I. plan broadly covers multiple issues without specific data or geographic precision, reading more as a general template applicable to any watershed. This report cites specific measurements and includes quantitative mass balance analysis showing that manure alone provides all crop P needs, and references fieldwork observations. On the other hand, the A.I. plan offers only generic statements without any quantification or referenced sources. The recommendations illustrate this contrast; this report provides specific actionable interventions such as the installation of structures at specific places, as well as proposing economically practical solutions to save farmers money and reduce P applications. The A.I. plan lists vague strategies like “promote cover cropping” and expand “green infrastructure” without specifying where and what type. This report is actionable as it tells decision-makers exactly where to intervene and what actions to take based on evidence, whereas the A.I. report produced a framework that doesn’t contain detailed specifications or references.

5. Recommendations

5.1. Sedimentation

5.1.1. Reducing Sediment Input into the Watershed

The largest and most problematic sediment source in the Fall Creek watershed is the Varna Cliffs. They continuously contribute to sedimentation, and their impact increases dramatically during storm periods. To limit the cliffs’ impact on water quality, any of the recommended actions can be implemented (or a combination):

- Installing engineered rock or wooden bank stabilization structures alongside the cliffs;
- Using landscape engineering techniques to naturally boost the integrity of the cliffs, such as planting willow trees with deep rooting that hold soil in place or installing geotextiles;
- Applying solid armor to particularly steep or unstable areas of the cliffs.

By adopting these measures, erosion from the Varna Cliffs can be minimized.

In Virgil Creek, the issues are similar, though the banks are less steep. To minimize the impacts of erosion in this stream, the following recommendations are made:

- Installing rocks or logs alongside the stream banks to minimize erosion during high flow;

- Adding vegetation such as willow trees and other native plants to reduce soil loss;
- Straightening the path of the stream to reduce meanders and subsequent erosion.

5.1.2. Improving the Monitoring Plan for Sedimentation

To improve the monitoring of sedimentation within the watershed, CSI should build on its current monitoring plan for the watershed. It can do so by adopting the following suggestions:

- Maintaining its current stations along Fall Creek and Virgil Creek, ensuring that equipment accurately measures sedimentation and turbidity during base and peak flows;
- Adding new measurement stations immediately upstream and downstream of the Varna Cliffs and meandering areas along Virgil Creek;
- Designing a robust, manual sampling plan for sedimentation immediately preceding and following major storm events in the watershed.
- Inspecting bank stabilization interventions for structural integrity and effectiveness seasonally and after major storms, replanting or repairing as needed;
- Reassessing erosion hotspots every three to five years as land use and stream geography change, and installing new bank stabilization structures as needed in the future.

By adopting these suggestions, CSI can create a more geographically and climatically robust monitoring system for sedimentation, capturing diverse conditions of the watershed.

5.2. Phosphorus

5.2.1. Reducing Phosphorous Input into the Watershed

Current recommendations to decrease the SRP are by the Agricultural Environmental Management (AEM) and Certified Nutrient Management plans made by the Soil and Water Conservation District. They are recommending decreasing the amount of fertilizer and manure that is applied to minimize soil and nutrient losses from farms to the streams. While this makes sense for reducing nutrient loading, asking farmers to reduce both inputs could impact crop yields and farm profitability. Participation by the farmers is also voluntary, which limits effectiveness if the measures are seen as economically harmful.

Based on the previous mass balance, it was found that farmers are already producing enough phosphorus in manure to meet their crop needs. The manure alone (641,850 kg) exceeded the total crop removal needs (635,250 kg) by 6,600 kg. This means that manure provides around 100% of crop P needs. The purchased fertilizer amount (260,700 kg) accounts for 41% of the crop removal necessary, meaning that an extra 41% of P is being applied to land unnecessarily.

Therefore, a more targeted measure would be to focus solely on eliminating purchased P fertilizer rather than reducing both manure and fertilizer applications. The recommendations are as follows:

- Maintaining current manure applications (which already meet crop P needs);
- Reducing watershed P surplus by eliminating or reducing the amount of fertilizer applied;
- Improving economic viability by reducing fertilizer costs while maintaining crop yields;
- Increasing voluntary participation through educating farms on the possibility to improve profitability while achieving environmental goals of reducing SRP in Fall Creek.

5.2.2. Improving the Monitoring Plan for Phosphorous

To enhance the Community Science Institute's effectiveness of SRP management efforts, the monitoring plan should incorporate farmer participation through incentive-based programs:

- Offering incentives, such as a rebate of a small amount of money for farmers to record and upload information related to their SRP usage, for amount of manure applied, P fertilizer applied, and what crops are being grown and their yield;
- Establishing payments where farmers receive compensation for documented reductions in P applications below current levels, incentivizing farmers to reduce the amount of P they are using.

6. Conclusions

Based on the analysis of sedimentation and phosphorous management in the Fall Creek watershed, a comprehensive approach addressing both pollutants is necessary to protect Cornell University's drinking water supply and improve overall water quality.

For sedimentation, the priority is stabilizing the Varna Cliffs through engineered structures, bioengineering techniques, and vegetation planting, as they represent the largest sediment source in the

watershed. Along Virgil Creek, bank stabilization measures using rocks, logs, and native vegetation should be implemented at meandering sections to reduce erosion during high flows. The monitoring plan should be enhanced by adding measurement stations upstream and downstream of critical erosion areas, developing storm event sampling protocols, and conducting seasonal inspections of all stabilization structures with reassessment of erosion hotspots every three to five years.

For phosphorus management, the focus should be on eliminating purchased fertilizer rather than reducing both manure and fertilizer applications, since the mass balance analysis shows that manure alone already provides enough phosphorus that the crops need. This targeted approach maintains current manure applications while reducing the watershed phosphorus surplus, saving farmers money on fertilizer costs and making voluntary participation more attractive. The monitoring plan should incorporate farmer participation through incentive-based programs, offering rebates for farmers who record and report their fertilizer and manure applications, and establishing payments for documented reductions in phosphorus use below current levels.

By implementing these measures, the Fall Creek watershed can achieve significant reductions in both sediment and phosphorus loading while improving farm economic viability and establishing a more robust monitoring system. The use of structural interventions, economic incentives, and enhanced monitoring can help create a better path towards more sustainable watershed management that protects both Cornell's drinking water supply and Cayuga Lake.

References

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Appendix A: Images



Figure 1: Areas of high bank erosion in Virgil Creek



Figure 2: visibly eroding area of Varna Cliffs along Fall Creek. Located between Freeville and Freese Road Bridge measurement sites.



Figure 3: area of Fall Creek appearing extremely shallow or with bare soil. Located between Freese Road Bridge and Freevile stations.

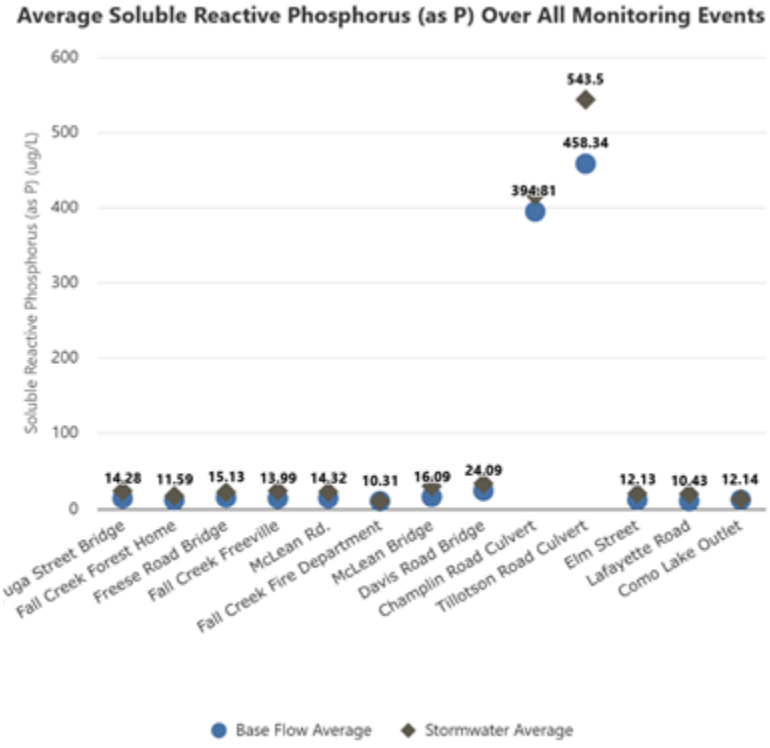


Figure 4. Average Soluble Reactive Phosphorous for Fall Creek

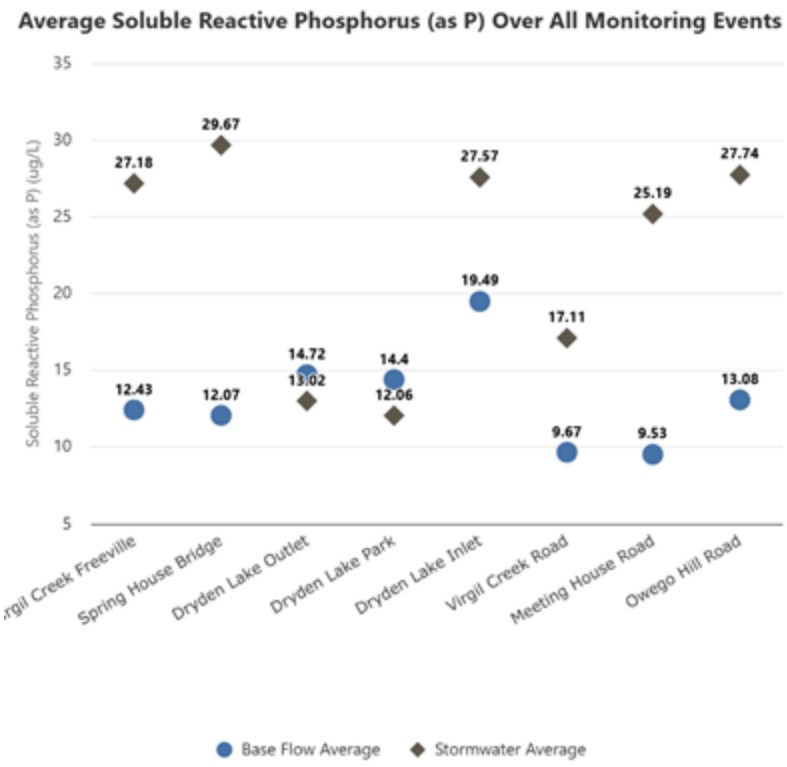


Figure 5. Average Soluble Reactive Phosphorous for Virgil Creek

Appendix B: Tables

Stream	% of bank erosion: base flow	% of bank erosion: high flow
Virgil Creek	55.0%	51.5%
Fall Creek	45.1%	96.7%

Table 1. Bank Erosion Values

County	P from recoverable livestock manure After Losses ($kg\ ha^{-1}$)	P from purchased fertilizer ($kg\ ha^{-1}$)	P crop removal ($kg\ ha^{-1}$)	P balance ($kg\ ha^{-1}$)	Average manure P application rate After Losses ($kg\ ha^{-1}$)	Average fertilizer P application rate ($kg\ ha^{-1}$)
Cayuga	25	13	23	17	95	19
Cortland	22	7	20	11	51	30
Tompkins	17	7	18	7	65	28

Table 2. Phosphorus Values from Godber et al. (2024)

Appendix C: AI Generated Report

Introduction

Fall Creek is one of the largest tributaries of Cayuga Lake, draining approximately 128 square miles of land across Cortland and Tompkins Counties before entering the lake at Ithaca, NY . The watershed encompasses a diverse landscape of forest, agriculture, and urban areas, supporting drinking water supply, fisheries, recreation, and ecological services. However, the system faces growing stressors including nutrient and sediment loading, habitat fragmentation, stormwater impacts from urban areas, and climate change. This management plan outlines a comprehensive strategy to protect and restore water quality, ecological integrity, and community benefits within the Fall Creek watershed.

Key Issues

1. **Water Quality and Sediment Loading**

Elevated nutrient inputs, especially phosphorus and nitrogen, from agricultural runoff and stormwater discharge. Sediment erosion from steep slopes and poorly managed construction sites contributing to turbidity in Cayuga Lake.

2. **Aquatic Habitat and Biodiversity**

Stream channelization, culverts, and barriers affecting fish passage (notably trout populations in upper tributaries). Declines in riparian habitat leading to warmer stream temperatures and reduced biodiversity.

3. **Urbanization and Stormwater**

Growth in Ithaca and surrounding communities increasing impervious surface area, accelerating runoff and flood risks. Untreated stormwater carrying pollutants directly into Fall Creek.

4. **Climate Change Resilience**

Increased frequency of extreme precipitation events leading to flooding, erosion, and infrastructure vulnerability. Rising water temperatures stressing cold-water fish habitats.

Goals

The watershed management plan is guided by four overarching goals:

1. **Protect and improve water quality** in Fall Creek and Cayuga Lake.
2. **Enhance ecological health and biodiversity** by restoring natural habitats and connectivity.
3. **Reduce flood risk and improve resilience** to climate change through nature-based solutions.
4. **Foster community engagement and stewardship** to ensure long-term watershed sustainability.

Recommended Strategies

1. Agricultural Best Management Practices (BMPs)

Promote cover cropping, riparian buffer strips, nutrient management planning, and reduced tillage. Support farmers through cost-share programs and technical assistance coordinated with local Soil and Water Conservation Districts.

2. Urban Stormwater Management

Expand green infrastructure in Ithaca, such as rain gardens, permeable pavements, and bioswales. Strengthen municipal stormwater ordinances to require on-site retention for new developments. Enhance maintenance of stormwater outfalls and monitoring of pollutant loads.

3. Riparian and Habitat Restoration

Restore riparian corridors with native vegetation to reduce bank erosion and improve shading. Identify and retrofit barriers (e.g., undersized culverts) to improve fish passage. Establish conservation easements along high-value stream reaches.

4. Erosion and Sediment Control

Enforce erosion-control practices at construction sites. Stabilize high-risk streambanks with bioengineering techniques. Coordinate with local highway departments to reduce roadside ditch erosion.

5. Climate Adaptation and Flood Resilience

Incorporate updated climate projections into watershed planning and zoning. Protect floodplains and wetlands as natural storage areas. Expand early warning and emergency preparedness systems for high-flow events.

6. Community Engagement and Education

Launch a watershed-wide stewardship campaign involving schools, residents, and local businesses. Develop citizen science programs for stream monitoring (temperature, turbidity, nutrient levels). Partner with Cornell University and local NGOs for ongoing research and adaptive management.

Implementation and Governance

The plan requires a collaborative governance model led by a **Fall Creek Watershed Partnership**, including municipalities, Tompkins County, Cornell University, New York State agencies, farmers, and local NGOs. Funding can be pursued through state and federal grants (e.g., NYS DEC Water Quality Improvement Program, EPA Section 319 grants), local cost-share initiatives, and university-community partnerships. A phased implementation is recommended:

Short-term (1–3 years): Establish governance framework, pilot BMP projects, enhance monitoring.
Medium-term (3–7 years): Expand riparian restoration, green infrastructure, and stormwater retrofits.
Long-term (7–15 years): Institutionalize adaptive management, track progress, and refine priorities.

Conclusion

Fall Creek is a critical ecological and community resource whose health directly influences the water quality of Cayuga Lake and the wellbeing of the Ithaca community. By implementing coordinated, science-based management actions and fostering strong partnerships, the watershed can become more resilient to climate change, support diverse ecosystems,